

Lecture 3: Proof Notes for Board Work

ISE 5405 – Optimization Theory

These notes contain the proofs to write on the board during lecture.

Green = Steps **Orange** = Key terms **Purple** = Results/Conclusions

1. Lemma (Closed Sets)

Statement

Suppose $S \subseteq \mathbb{R}^n$ is a **closed** set. Then:

S contains all its boundary points: $\partial S \subseteq S$

Its complement $\mathbb{R}^n \setminus S$ is **open**

For any **convergent sequence** $\{\mathbf{x}_k\}$ in S with $\lim_{k \rightarrow \infty} \mathbf{x}_k = \bar{\mathbf{x}}$, we have $\bar{\mathbf{x}} \in S$

Step 1: Prove $\partial S \subseteq S$

[Let's take an arbitrary boundary point and show it must be in S]

Let $\mathbf{x} \in \partial S$

By **definition of boundary**:

for every $\epsilon > 0$, $\mathcal{N}_\epsilon(\mathbf{x})$ contains at least one point in S

$\Rightarrow S \cap \mathcal{N}_\epsilon(\mathbf{x}) \neq \emptyset$ for all $\epsilon > 0$

By **definition of closure**: $\mathbf{x} \in \text{cl}(S)$

Since S is **closed**: $S = \text{cl}(S)$

Therefore $\mathbf{x} \in S$ $\square$

Step 2: Prove $\mathbb{R}^n \setminus S$ is open

[Show every point in the complement has a neighborhood in the complement]

Let $\mathbf{x} \in \mathbb{R}^n \setminus S$ (i.e., $\mathbf{x} \notin S$)

Want to show: $\exists \epsilon > 0$ such that $\mathcal{N}_\epsilon(\mathbf{x}) \subseteq \mathbb{R}^n \setminus S$

[Use proof by contradiction]

Suppose not : for all $\epsilon > 0$, $\mathcal{N}_\epsilon(\mathbf{x}) \cap S \neq \emptyset$

$\Rightarrow$ By definition of closure: $\mathbf{x} \in \text{cl}(S)$

Since S is closed: $\text{cl}(S) = S$

$\Rightarrow \mathbf{x} \in S$ **Contradiction!**

Therefore such $\epsilon > 0$ exists, so $\mathbb{R}^n \setminus S$ is open $\square$

Step 3: Prove limits of sequences in S stay in S

Let $\{\mathbf{x}_k\}$ be a sequence in S with $\mathbf{x}_k \rightarrow \bar{\mathbf{x}}$

By **definition of convergence**:

For any $\epsilon > 0$: $\exists K$ such that $\|\mathbf{x}_k - \bar{\mathbf{x}}\| < \epsilon$ for all $k \geq K$

$\Rightarrow \mathbf{x}_k \in \mathcal{N}_\epsilon(\bar{\mathbf{x}})$ for $k \geq K$

Since $\mathbf{x}_k \in S$: we have $S \cap \mathcal{N}_\epsilon(\bar{\mathbf{x}}) \neq \emptyset$

This holds for **all** $\epsilon > 0$

$\Rightarrow$ By definition of closure: $\bar{\mathbf{x}} \in \text{cl}(S)$

Since S is closed: $\bar{\mathbf{x}} \in S$

Therefore $\bar{\mathbf{x}} \in S$ $\square$

2. Lemma (Open Sets)

Statement

A set $S \subseteq \mathbb{R}^n$ is **open** if and only if it does not contain any of its boundary points:

$$\partial S \cap S = \emptyset$$

Step 1: $(\Rightarrow)$ Assume S is open. Show $\partial S \cap S = \emptyset$

[Show no point of S can be a boundary point]

Let $\mathbf{x} \in S$

Since S is **open**: $S = \text{int}(S)$

$\Rightarrow \mathbf{x} \in \text{int}(S)$

By **definition of interior**: $\exists \epsilon > 0$ with $\mathcal{N}_\epsilon(\mathbf{x}) \subseteq S$

$\Rightarrow \mathcal{N}_\epsilon(\mathbf{x})$ contains **no points outside** S

By **definition of boundary**: $\mathbf{x} \notin \partial S$

No point of S is a boundary point: $\partial S \cap S = \emptyset$ $\square$

Step 2: $(\Leftarrow)$ Assume $\partial S \cap S = \emptyset$. Show S is open

[Show every point of S is an interior point]

Let $\mathbf{x} \in S$

Since $\partial S \cap S = \emptyset$: we have $\mathbf{x} \notin \partial S$

Also $\mathbf{x} \in S$, so $\mathbf{x} \in \text{cl}(S)$ *[points in S are in its closure]*

Key fact: $\text{cl}(S) = \text{int}(S) \cup \partial S$

Since $\mathbf{x} \in \text{cl}(S)$ and $\mathbf{x} \notin \partial S$:

$\Rightarrow \mathbf{x} \in \text{int}(S)$

This holds for all $\mathbf{x} \in S$, so $S \subseteq \text{int}(S)$

But always $\text{int}(S) \subseteq S$

Therefore $S = \text{int}(S)$, so S is open $\square$

3. Lemma (Boundary)

Statement

The boundary of a set $S \subseteq \mathbb{R}^n$ is given by:

$$\partial S = \text{cl}(S) \setminus \text{int}(S)$$

Step 1: $(\subseteq)$ Show $\partial S \subseteq \text{cl}(S) \setminus \text{int}(S)$

Let $\mathbf{x} \in \partial S$

(a) Show $\mathbf{x} \in \text{cl}(S)$

By **def of boundary**: for every $\epsilon > 0$, $\mathcal{N}_\epsilon(\mathbf{x})$ contains ≥ 1 point in S

$\Rightarrow S \cap \mathcal{N}_\epsilon(\mathbf{x}) \neq \emptyset$ for all $\epsilon > 0$

By **def of closure**: $\mathbf{x} \in \text{cl}(S)$ ✓

(b) Show $\mathbf{x} \notin \text{int}(S)$

By **def of boundary**: for every $\epsilon > 0$, $\mathcal{N}_\epsilon(\mathbf{x})$ contains ≥ 1 point **not** in S

$\Rightarrow$ There is **no** $\epsilon > 0$ with $\mathcal{N}_\epsilon(\mathbf{x}) \subseteq S$

By **def of interior**: $\mathbf{x} \notin \text{int}(S)$ ✓

Therefore $\mathbf{x} \in \text{cl}(S) \setminus \text{int}(S)$ □

Step 2: $(\supseteq)$ Show $\text{cl}(S) \setminus \text{int}(S) \subseteq \partial S$

Let $\mathbf{x} \in \text{cl}(S) \setminus \text{int}(S)$

(a) From $\mathbf{x} \in \text{cl}(S)$:

For every $\epsilon > 0$: $S \cap \mathcal{N}_\epsilon(\mathbf{x}) \neq \emptyset$

$\Rightarrow$ Every neighborhood contains a point **in** S ✓

(b) From $\mathbf{x} \notin \text{int}(S)$:

For every $\epsilon > 0$: $\mathcal{N}_\epsilon(\mathbf{x}) \not\subseteq S$

$\Rightarrow$ Every neighborhood contains a point **not in** S ✓

By **definition of boundary**: $\mathbf{x} \in \partial S$

Therefore $\mathbf{x} \in \partial S$ □

4. Bolzano–Weierstrass Theorem

Statement

Every **bounded** sequence in $\mathbb{R}^n$ has a **convergent subsequence**.

[We'll prove this for $n = 1$ using the bisection method. The general case follows by applying this to each coordinate.]

Step 1: Setup

Let $\{x_k\}$ be a bounded sequence in $\mathbb{R}$

Since bounded: $\exists M$ such that $|x_k| \leq M$ for all k

$\Rightarrow$ All terms lie in $[a_0, b_0] = [-M, M]$

Step 2: Bisection construction

[We'll build nested intervals, each containing infinitely many terms]

Bisect $[a_0, b_0]$ into two halves

At least one half contains **infinitely many** terms of $\{x_k\}$

Call this half $[a_1, b_1]$ (length = M)

Pick $x_{k_1} \in [a_1, b_1]$

Repeat : Bisect $[a_1, b_1]$, choose half with infinitely many terms

Call it $[a_2, b_2]$ (length = $M/2$)

Pick $x_{k_2} \in [a_2, b_2]$ with $k_2 > k_1$

Continue : Get nested intervals $[a_j, b_j]$ with length $\frac{2M}{2^j} \rightarrow 0$

Pick $x_{k_j} \in [a_j, b_j]$ with $k_1 < k_2 < k_3 < \dots$

Step 3: Identify the limit

By **Nested Interval Theorem**:

$$\Rightarrow \bigcap_{j=1}^{\infty} [a_j, b_j] = \{\bar{x}\} \text{ for some } \bar{x} \in \mathbb{R}$$

Step 4: Show convergence

For any $\epsilon > 0$: choose J large enough that $\frac{2M}{2^J} < \epsilon$

For $j \geq J$: $x_{k_j} \in [a_j, b_j]$ and $\bar{x} \in [a_j, b_j]$

$$\Rightarrow |x_{k_j} - \bar{x}| \leq b_j - a_j = \frac{2M}{2^j} < \epsilon$$

Therefore $x_{k_j} \rightarrow \bar{x}$ $\square$

[For $\mathbb{R}^n$: apply to each coordinate, use diagonal argument to get common subsequence]

5. Weierstrass' Theorem

Statement

Let S be a nonempty **compact** set, and let $f : S \rightarrow \mathbb{R}$ be **continuous** on S .
Then $\min\{f(\mathbf{x}) : \mathbf{x} \in S\}$ **attains its minimum**.

Step 1: Show infimum exists and is finite

Let $\gamma = \inf\{f(\mathbf{x}) : \mathbf{x} \in S\}$

Since $S \neq \emptyset$: γ exists in $[-\infty, \infty)$

[We need to show $\gamma > -\infty$ and is attained]

Step 2: Construct minimizing sequence

By **definition of infimum**:

For each $k \in \mathbb{N}$, $\exists \mathbf{x}_k \in S$ such that:

$$\gamma \leq f(\mathbf{x}_k) < \gamma + \frac{1}{k}$$

$$\Rightarrow f(\mathbf{x}_k) \rightarrow \gamma \text{ as } k \rightarrow \infty$$

Step 3: Extract convergent subsequence

$\{\mathbf{x}_k\}$ is a sequence in S

S is **bounded** (since compact)

$$\Rightarrow \{\mathbf{x}_k\} \text{ is bounded}$$

By **Bolzano–Weierstrass**: $\exists$ convergent subsequence $\mathbf{x}_{k_j} \rightarrow \bar{\mathbf{x}}$

Step 4: Show limit is in S

S is **closed** (since compact)

$\{\mathbf{x}_{k_j}\}$ is a sequence in S converging to $\bar{\mathbf{x}}$

By **closed set property**: $\bar{\mathbf{x}} \in S$ ✓

Step 5: Show $f(\bar{\mathbf{x}}) = \gamma$

f is **continuous** at $\bar{\mathbf{x}}$

$$\mathbf{x}_{k_j} \rightarrow \bar{\mathbf{x}}$$

$$\Rightarrow f(\mathbf{x}_{k_j}) \rightarrow f(\bar{\mathbf{x}}) \quad [\text{continuity preserves limits}]$$

But we know $f(\mathbf{x}_k) \rightarrow \gamma$, so $f(\mathbf{x}_{k_j}) \rightarrow \gamma$

By **uniqueness of limits**: $f(\bar{\mathbf{x}}) = \gamma$

Step 6: Conclude

$\bar{\mathbf{x}} \in S$ and $f(\bar{\mathbf{x}}) = \gamma = \inf\{f(\mathbf{x}) : \mathbf{x} \in S\}$

$\Rightarrow \gamma > -\infty$ (since $f(\bar{\mathbf{x}})$ is a real number)

Therefore $\bar{\mathbf{x}}$ is a minimizer: $f(\bar{\mathbf{x}}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in S$ $\square$

6. Closest-Point Theorem

Statement

Given a nonempty **closed convex** set $S \subseteq \mathbb{R}^n$ and a point $\mathbf{y} \notin S$:

- (1) There exists a **unique** point $\bar{\mathbf{x}} \in S$ with minimum distance from $\mathbf{y}$
- (2) $\bar{\mathbf{x}}$ is the minimizing point **if and only if** $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$

Step 1: Prove existence of a closest point

[Use Weierstrass' theorem on a bounded subset]

Define $f(\mathbf{x}) = \|\mathbf{y} - \mathbf{x}\|^2$ *[distance squared, continuous]*

Problem: S may be unbounded, can't apply Weierstrass directly

[Fix this by restricting to a bounded subset]

Pick any $\mathbf{x}_0 \in S$ *[exists since $S \neq \emptyset$]*

Let $r = \|\mathbf{y} - \mathbf{x}_0\|$ *[distance from $\mathbf{y}$ to $\mathbf{x}_0$]*

Define $S' = S \cap \overline{B}(\mathbf{y}, r)$ *[intersection with closed ball]*

Claim: S' is nonempty, closed, bounded (hence compact)

- Nonempty: $\mathbf{x}_0 \in S'$ ✓
- Closed: intersection of closed sets ✓
- Bounded: subset of $\overline{B}(\mathbf{y}, r)$ ✓

By **Weierstrass' Theorem**: $\exists \bar{\mathbf{x}} \in S'$ minimizing f over S'

Claim: $\bar{\mathbf{x}}$ also minimizes f over all of S

For any $\mathbf{x} \in S \setminus S'$: $\|\mathbf{y} - \mathbf{x}\| > r \geq \|\mathbf{y} - \bar{\mathbf{x}}\|$

Therefore $\bar{\mathbf{x}}$ is the closest point in S to $\mathbf{y}$ $\square$

Step 2: Prove uniqueness

[Use strict convexity of the squared distance function]

Suppose $\bar{\mathbf{x}}_1$ and $\bar{\mathbf{x}}_2$ are both closest to $\mathbf{y}$

Let $d = \|\mathbf{y} - \bar{\mathbf{x}}_1\| = \|\mathbf{y} - \bar{\mathbf{x}}_2\|$ *[same minimum distance]*

Consider $\hat{\mathbf{x}} = \frac{1}{2}\bar{\mathbf{x}}_1 + \frac{1}{2}\bar{\mathbf{x}}_2$

Since S is **convex**: $\hat{\mathbf{x}} \in S$

Compute distance from $\mathbf{y}$ to $\hat{\mathbf{x}}$:

$$\begin{aligned}\|\mathbf{y} - \hat{\mathbf{x}}\|^2 &= \left\| \mathbf{y} - \frac{\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2}{2} \right\|^2 \\ &= \left\| \frac{(\mathbf{y} - \bar{\mathbf{x}}_1) + (\mathbf{y} - \bar{\mathbf{x}}_2)}{2} \right\|^2\end{aligned}$$

By **strict convexity of $\|\cdot\|^2$** :

$$\|\mathbf{y} - \hat{\mathbf{x}}\|^2 < \frac{1}{2}\|\mathbf{y} - \bar{\mathbf{x}}_1\|^2 + \frac{1}{2}\|\mathbf{y} - \bar{\mathbf{x}}_2\|^2 = d^2$$

$$\Rightarrow \|\mathbf{y} - \hat{\mathbf{x}}\| < d \quad \textbf{Contradiction!} \quad (\text{unless } \bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_2)$$

Therefore the closest point is unique $\square$

Step 3: $(\Rightarrow)$ If $\bar{\mathbf{x}}$ minimizes distance, show $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$

Let $\bar{\mathbf{x}}$ be the closest point in S to $\mathbf{y}$

Let $\mathbf{x} \in S$ be arbitrary

For $\lambda \in [0, 1]$: $\mathbf{x}_\lambda = \bar{\mathbf{x}} + \lambda(\mathbf{x} - \bar{\mathbf{x}}) = (1 - \lambda)\bar{\mathbf{x}} + \lambda\mathbf{x}$

Since S is **convex**: $\mathbf{x}_\lambda \in S$

Since $\bar{\mathbf{x}}$ minimizes distance:

$$\|\mathbf{y} - \bar{\mathbf{x}}\|^2 \leq \|\mathbf{y} - \mathbf{x}_\lambda\|^2 \text{ for all } \lambda \in [0, 1]$$

Expand right side:

$$\begin{aligned} \|\mathbf{y} - \mathbf{x}_\lambda\|^2 &= \|\mathbf{y} - \bar{\mathbf{x}} - \lambda(\mathbf{x} - \bar{\mathbf{x}})\|^2 \\ &= \|\mathbf{y} - \bar{\mathbf{x}}\|^2 - 2\lambda(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) + \lambda^2\|\mathbf{x} - \bar{\mathbf{x}}\|^2 \end{aligned}$$

$$\Rightarrow 0 \leq -2\lambda(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) + \lambda^2\|\mathbf{x} - \bar{\mathbf{x}}\|^2$$

Divide by $\lambda > 0$:

$$0 \leq -2(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) + \lambda\|\mathbf{x} - \bar{\mathbf{x}}\|^2$$

Take $\lambda \rightarrow 0^+$:

$$0 \leq -2(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}})$$

Therefore $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ $\square$

Step 4: $(\Leftarrow)$ If $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$, show $\bar{\mathbf{x}}$ minimizes

Assume $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$

For any $\mathbf{x} \in S$:

$$\begin{aligned} \|\mathbf{y} - \mathbf{x}\|^2 &= \|(\mathbf{y} - \bar{\mathbf{x}}) - (\mathbf{x} - \bar{\mathbf{x}})\|^2 \\ &= \|\mathbf{y} - \bar{\mathbf{x}}\|^2 - 2(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) + \|\mathbf{x} - \bar{\mathbf{x}}\|^2 \end{aligned}$$

Since $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$:

$$\Rightarrow -2(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \geq 0$$

Also $\|\mathbf{x} - \bar{\mathbf{x}}\|^2 \geq 0$

$$\Rightarrow \|\mathbf{y} - \mathbf{x}\|^2 \geq \|\mathbf{y} - \bar{\mathbf{x}}\|^2$$

$$\Rightarrow \|\mathbf{y} - \mathbf{x}\| \geq \|\mathbf{y} - \bar{\mathbf{x}}\|$$

Therefore $\bar{\mathbf{x}}$ minimizes distance to $\mathbf{y}$ over S $\square$

7. Separating Hyperplane Theorem

Statement

Let S be a nonempty **closed convex** set in $\mathbb{R}^n$ and $\mathbf{y} \notin S$.
Then $\exists$ nonzero $\mathbf{p}$ and scalar α such that:

$$\mathbf{p}^T \mathbf{y} > \alpha \quad \text{and} \quad \mathbf{p}^T \mathbf{x} \leq \alpha \quad \text{for each } \mathbf{x} \in S$$

Step 1: Apply Closest-Point Theorem

Since S is nonempty, **closed**, and **convex**, and $\mathbf{y} \notin S$:

By **Closest-Point Theorem**:

$\exists$ unique $\bar{\mathbf{x}} \in S$ closest to $\mathbf{y}$

Moreover: $(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$ for all $\mathbf{x} \in S$

Step 2: Define the separating hyperplane

Set $\mathbf{p} = \mathbf{y} - \bar{\mathbf{x}}$

[Note: $\mathbf{p} \neq \mathbf{0}$ since $\mathbf{y} \neq \bar{\mathbf{x}}$ (because $\mathbf{y} \notin S$ but $\bar{\mathbf{x}} \in S$)]

Set $\alpha = \mathbf{p}^T \bar{\mathbf{x}} = (\mathbf{y} - \bar{\mathbf{x}})^T \bar{\mathbf{x}}$

Step 3: Verify $\mathbf{p}^T \mathbf{x} \leq \alpha$ for all $\mathbf{x} \in S$

From Closest-Point condition:

$$(\mathbf{y} - \bar{\mathbf{x}})^T(\mathbf{x} - \bar{\mathbf{x}}) \leq 0$$

$$\Rightarrow \mathbf{p}^T \mathbf{x} - \mathbf{p}^T \bar{\mathbf{x}} \leq 0$$

$$\Rightarrow \mathbf{p}^T \mathbf{x} \leq \mathbf{p}^T \bar{\mathbf{x}} = \alpha \quad \checkmark$$

Step 4: Verify $\mathbf{p}^T \mathbf{y} > \alpha$

Compute $\mathbf{p}^T \mathbf{y}$:

$$\begin{aligned} \mathbf{p}^T \mathbf{y} &= (\mathbf{y} - \bar{\mathbf{x}})^T \mathbf{y} \\ &= (\mathbf{y} - \bar{\mathbf{x}})^T \bar{\mathbf{x}} + (\mathbf{y} - \bar{\mathbf{x}})^T (\mathbf{y} - \bar{\mathbf{x}}) \\ &= \alpha + \|\mathbf{y} - \bar{\mathbf{x}}\|^2 \end{aligned}$$

Since $\mathbf{y} \neq \bar{\mathbf{x}}$: $\|\mathbf{y} - \bar{\mathbf{x}}\|^2 > 0$

$$\Rightarrow \mathbf{p}^T \mathbf{y} > \alpha \quad \checkmark$$

This completes the proof $\square$